OPTION PRICING · PYTHON PROJECT

# Pricing options with the Fourier transform

*Every strike at once, from the model's fingerprint*

Padraig Middleton · October 2026 · github.com/poggey/fourier-option-pricing

**4,096**
strikes priced in one FFT pass

**0.2 ms**
for every strike, over 9,000× faster than one at a time

**$6 \times 10^{-14}$**
largest gap from Black-Scholes

I built an option pricer that only needs a model's *fingerprint* (the Fourier transform of where the share might end up), then checked it against Black-Scholes, where the right answer is known.

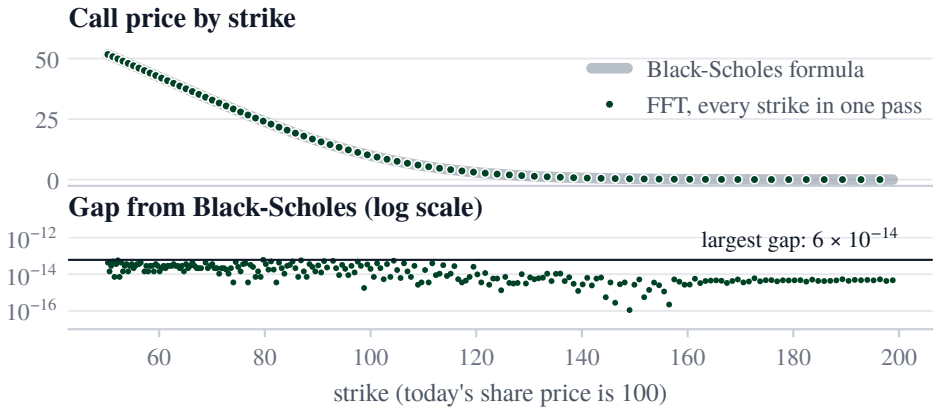



**Figure 1.** One FFT pass against the Black-Scholes formula, at the 225 grid strikes from 50 to 200.

## 01 The idea

*An option's price is an average over everywhere the share could end up.*

- For realistic models those chances have no formula, but their fingerprint usually does.
- So I price options from the fingerprint alone (Carr and Madan, 1999), with the FFT doing every strike at once.
- Why it matters: banks re-price thousands of options every time they fit a model to market prices.

## 02 Notation

*Everything is measured in logs, because shares move in percentages.*

- A 10% rise and a 10% fall become equal steps, so the outcomes form a symmetric bell curve.
- Running example: share at 100, strike 100, one year, 4% interest, 20% volatility. Its exact Black-Scholes price is 9.925054.

| Symbol | Plain name | The idea | Example |
|---|---|---|---|
| $S_0$ | Today's share price | Where the share starts | 100 |
| $K$ | Strike | The fixed price you can buy at | 100 |
| $T$, $r$ | Time, interest rate | Years until the option ends; what money earns meanwhile | 1, 4% |
| vol | Volatility | How much the share typically moves in a year | 20% |
| $s$, $k$ | Log final price, log strike | The final price and the strike on the percentage scale | $k = \ln 100 = 4.605$ |
| $q(s)$ | Probability curve | How likely each log final price is | a bell curve |
| $C(k)$, $D(k)$ | Call price, damped price | What the option is worth at each strike; the same, damped | 9.925 |
| $a$ | Damping strength | How hard the left end of the price curve is pushed down | 1.5 |
| $u$, $v$ | Spin speeds | How fast the arrows turn: $u$ for chances, $v$ for prices | |
| $F(u)$, $W(v)$ | Fingerprint, wound-up price | The average arrow; the damped price in frequency world | |

## 03 The price is an average

*Fair price = average payoff, weighted by how likely each outcome is, then discounted.*

- A call pays $\max(S_T - K, 0)$: with strike 100, ending at 120 pays 20 and ending at 90 pays nothing.
- Toy version: the share ends at 80, 100 or 120 (25%, 50%, 25%). Only 120 pays (20), so the price is $0.25 \times 20 \times e^{-0.04} \approx 4.80$.
- Real version: the outcomes become a curve and the sum becomes an integral.

$$C(k) = e^{-rT} \int_k^\infty \left(e^s - e^k\right) q(s)\, ds$$

- Under Black-Scholes, $q$ is a bell curve with centre $m = \ln S_0 + (r - \frac{1}{2}\text{vol}^2)T = 4.6252$ and spread $\text{vol}\sqrt{T} = 0.20$.
- The height is chance *per unit width*: 2.0 at the centre means about a 2% chance of ending within 1%.

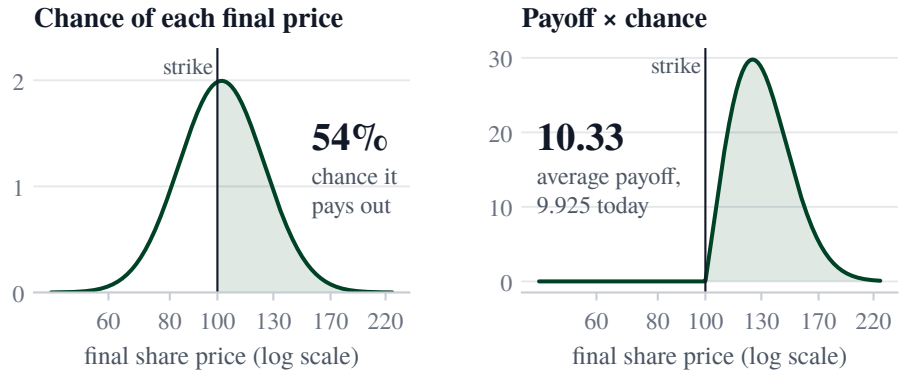



**Figure 2.** Shaded on the left: the chance it pays out, 54% ($N(d_2)$ in Black-Scholes). On the right: the average payoff, 10.33.

## 04 The fingerprint

*Turn every outcome into an arrow, weight it by its chance, and average. That's the fingerprint.*

- $e^{ix} = \cos x + i \sin x$ is an arrow of length 1 at angle $x$. At spin speed $u$, outcome $s$ points at angle $us$.
- Writing each outcome as centre plus deviation, $s = m + x$, the average splits into two parts:

$$F(u) = \int e^{ius}\, q(s)\, ds = \underbrace{e^{ium}}_{\text{rotation}} \times \underbrace{e^{-\frac{1}{2}\text{vol}^2 T u^2}}_{\text{shrink}}$$

- Rotation: where the outcomes are. Shrink: how spread out they are, as the arrows fan out and cancel.
- The payoff isn't in it, so a new model means changing only this one function.

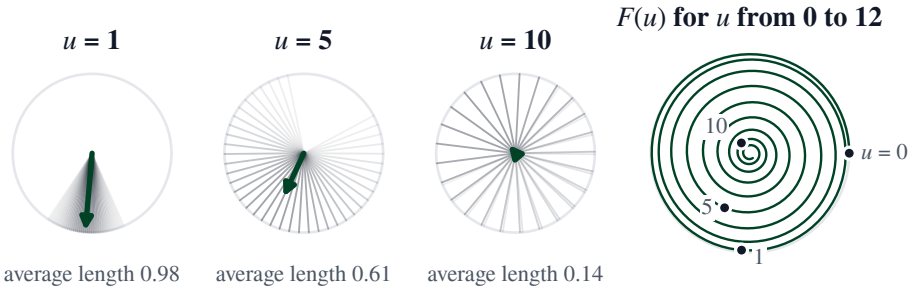



**Figure 3.** Grey: each outcome's arrow. Green: their average, which shrinks and spirals in as the spin speeds up.

**The plan.** I only know the share through its fingerprint, which lives in frequency world. So I move the call price there, work it out, and bring it back in four steps.

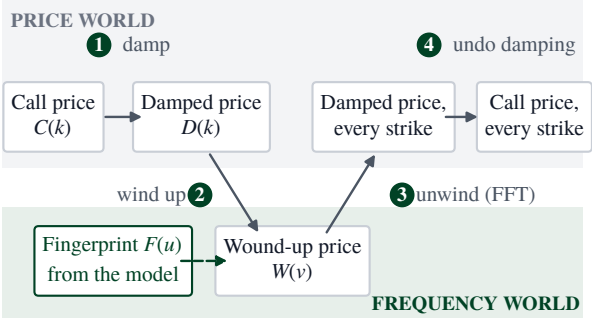



**Figure 4.** Step 2 never needs any prices: it's built from the fingerprint.

## 05 The snag: damping

*The call price never dies away on the left, so I push it down before transforming it.*

- Right end: the price falls to zero, since the share would have to double to pass strike 200.
- Left end: buying the share for almost nothing is worth about 100, however small the strike.
- Fix: multiply by $K^a$, which is tiny on the left. Strike 0.0001 goes from 99.9999 to 0.0001.

$$D(k) = e^{ak}\, C(k) = K^a \times C$$

- Nothing is lost: multiplying by $e^{-ak}$ at the end undoes it.

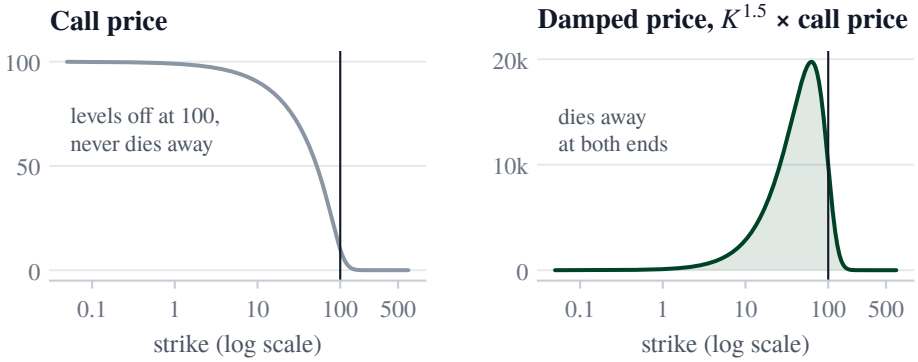



**Figure 5.** Left: levels off at 100. Right: damped by $K^{1.5}$, it dies away at both ends.

## 06 Winding it up

*Swap the order: wind up each outcome across strikes, then average. That needs only the payoff and the fingerprint.*

A small version first: 3 outcomes, 3 strikes, damping 1.5. Each cell is chance × $K^{1.5}$ × payoff.

| Strike / outcome | 80 (25%) | 100 (50%) | 120 (25%) | Row total |
|---|---|---|---|---|
| 70 | 1,464 | 8,785 | 7,321 | **17,570** |
| 90 | 0 | 4,269 | 6,404 | **10,673** |
| 110 | 0 | 0 | 2,884 | **2,884** |
| Column total | **1,464** | **13,054** | **16,609** | 31,127 |

- **Rows** add up to damped prices. That needs the price at every strike, which I don't have.
- **Columns** add up to one outcome's share. That needs only the payoff and its chance.
- Same grand total either way: $-23{,}141 + 18{,}532i$ here at $v = 2$, and 30,520.88 for the full curve.
- Each column has a closed form, so the whole table collapses to one formula:

$$W(v) = \frac{e^{-rT}\, F(v - (a+1)i)}{a^2 + a - v^2 + i(2a+1)v}$$

- Discounting, times the chances (fingerprint at a shifted speed), over the payoff's shape. Damping must be positive, or the bottom hits zero at $v = 0$.

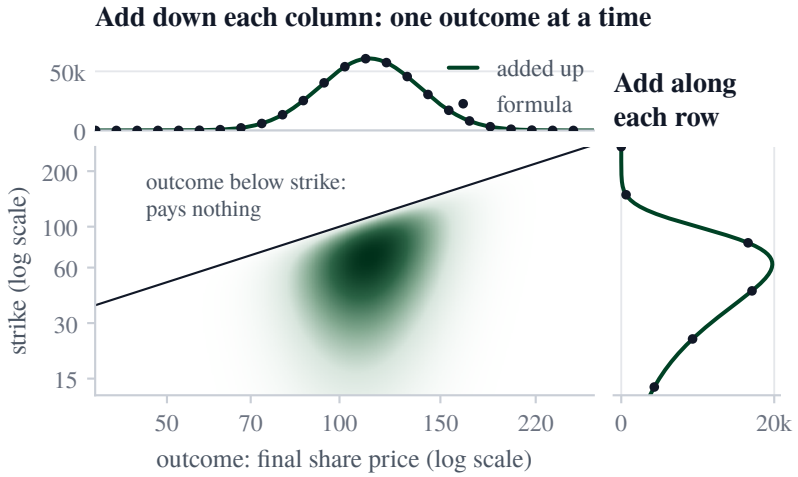



**Figure 6.** The full table at $v = 0$. Columns match the formula, rows match the damped prices; both total 30,520.88.

## 07 Unwinding with the FFT

*Unwind back to strikes and undo the damping. The FFT does every strike at once.*

$$C(k) = \frac{e^{-ak}}{\pi} \int_0^\infty \mathrm{Re}\left[e^{-ivk}\, W(v)\right] dv$$

- Why $\frac{1}{\pi}$: the full unwind has $\frac{1}{2\pi}$, but prices are real, so $+v$ and $-v$ pair up and only half is needed.
- Strike 100: the integral is about 31,180, and $31{,}180/(100^{1.5}\pi) = 9.925$.
- If (gap between spin speeds) × (gap between log strikes) $= 2\pi/N$, the sum is exactly an FFT.
- For 4,096 strikes that's about 49,000 steps instead of 16.8 million, with strikes 0.6% apart.

## 08 Results

*It matches Black-Scholes to 12 decimal places, over 9,000 times faster than one strike at a time.*

- Accuracy: largest gap $6 \times 10^{-14}$ across the 225 strikes from 50 to 200 (Figure 1).
- Single strikes: 23.906164, 9.925054 and 2.999949 at strikes 80, 100 and 120. Damping 1.0 gives the same.
- Speed, on my laptop: 0.2 ms for all 4,096 strikes by FFT, against 0.5 ms *per strike* one at a time.

## 09 Limits and next steps

*Two settings decide how accurate it is. Next, I want a model where shares can crash.*

- Damping: the gap is $6 \times 10^{-14}$ at 1.5, but $1 \times 10^{-9}$ at 1.0 and $4 \times 10^{-4}$ at 0.5.
- Grid: spin speeds 1.0 apart put strikes 0.15% apart, but the gap grows to $8 \times 10^{-3}$.
- European options only: one payoff, on one fixed date.
- **Next:** a jump model (Merton's) needs only a new fingerprint, and should show why crash insurance has cost more than Black-Scholes says since 1987. Then fit it to real market prices.